

Chapter 2: Why Do Companies Ask DSA?

Introduction

Many students believe that companies ask Data Structures and Algorithms (DSA) questions only to test coding skills. However, that is not the main reason. Companies are more interested in understanding how candidates think, analyze problems, and create efficient solutions.

What Companies Want to Know

When you solve DSA problems, companies can evaluate:

- Can you think logically?
- Can you break a complex problem into smaller steps?
- Can you find efficient solutions?
- Can you optimize your code?
- Can you work with large amounts of data?
- Can you handle real-world software challenges?

A person who can write code is valuable, but a person who can solve problems efficiently is even more valuable.

The Real Challenge: Scale

Consider a website that stores student records.

10 Students → Easy
100 Students → Easy
1,000 Students → Easy
10,000 Students → Manageable
1,00,000 Students → Challenging
10,00,000+ Students → Difficult

For small amounts of data, almost any solution works. However, when millions of users start using an application, inefficient solutions become slow and expensive.

Companies ask:

"Will your solution still work efficiently when the data becomes huge?"

This is where DSA becomes important.

Real-Life Example: Library Search

Imagine a library containing 1,00,000 books.

Method 1: Unorganized Search (Bad Approach)

1. Check the first book.
2. If it is not the correct book, check the next one.
3. Continue until the book is found.

Problems:

- Takes a long time.
- Wastes effort.
- Becomes slower as the library grows.

Method 2: Organized Search (Good Approach)

1. Go directly to the correct alphabetical section.
2. Narrow down the search area.
3. Find the book quickly.

Benefits:

- Much faster.
- Less work.
- Scales well for large collections.

This is exactly what DSA teaches.

How DSA Helps in Software Development

Every popular application uses DSA behind the scenes.

Social Media

- Finding friends quickly
- Showing posts efficiently
- Searching users instantly

E-Commerce Websites

- Product searching
- Product recommendations
- Order management

Navigation Apps

- Finding shortest routes
- Traffic optimization
- Route calculations

Banking Systems

- Fast transaction processing
- Secure data management
- Efficient record searching

Example of Optimization

Suppose you need to find a student named Rahul in a list of 1,00,000 students.

Approach 1: Check every student one by one.

Time Required: High

Approach 2: Use an organized data structure and an efficient searching algorithm.

Time Required: Very Low

Both approaches give the same answer. The difference is speed.

Why DSA Is Important for Interviews

DSA helps interviewers evaluate:

- Problem-solving ability
- Logical thinking
- Analytical skills
- Optimization skills
- Coding fundamentals
- Understanding of computer science concepts

Key Takeaways

- DSA is not just about coding.
- Companies use DSA to test problem-solving skills.
- Efficient solutions matter when data becomes very large.
- Good algorithms save time and resources.
- Good data structures organize information effectively.
- Modern software systems rely heavily on DSA.
- Strong DSA skills increase interview success.

Remember:

"Anyone can write code that works for 10 users.
A software engineer writes code that works for 10 million users."

That is why companies ask DSA.